Project Proposal: Water Budget Calculator

Hyper-Local Precipitation vs Evapotranspiration for a Garden in Falenty Nowe, Poland

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1. Websites to Scrape

#	Website	URL	Tool Used
1	IMGW Public Data Archive	danepubliczne.imgw.pl/data/ dane_pomiarowo_obserwacyjne/	**Scrapy** + **Regex**
2	Open-Meteo Historical API	archive-api.open-meteo.com/v1/archive	**Requests**
3	Meteostat Station Page	meteostat.net/en/station/12375	**Selenium** + **BS4**

2. Legal Proof

IMGW (danepubliczne.imgw.pl): Data is published under the Polish Open Data Policy (ustawa z dnia 11 sierpnia 2021 r. o otwartych danych, Dz.U. 2023 poz. 1524) and EU regulation 2023/138. The regulations state: *"Korzystający może używać nieodpłatnie udostępnionych danych do celów prywatnych"* (free for private/educational use). The robots.txt does not disallow access to the /data/ directory.

Open-Meteo (open-meteo.com): Free JSON API for non-commercial use. No API key required. Data licensed under CC BY 4.0 (Attribution 4.0 International). The API is explicitly designed for programmatic access — no robots.txt restrictions on API endpoints.

Meteostat (meteostat.net): The robots.txt explicitly allows all crawlers (Allow: /). Data is provided under CC BY-NC 4.0. The project is open-source (MIT license). This is a non-commercial educational project.

3. Elements to Scrape

The goal is to build a **Water Budget Model** for a garden in Falenty Nowe (near Warsaw) by computing the daily water balance: **$P - ET_0$** (Precipitation minus Evapotranspiration). The final output is a structured DataFrame with 462+ daily rows (Jan 2025 – Apr 2026).

Source	Elements Scraped	Method
IMGW	Daily CSV/ZIP files: temperature (min/max/avg), precipitation sum (mm), sunshine hours, snow depth. Station: Warszawa-Okecie (ID 352200500, WMO 12375).	**Scrapy** crawls Apache directory index, follows year links, downloads ZIPs. **Regex** extracts numeric values from CSV rows.
Open-Meteo	Daily JSON data: temperature, precipitation, rain, snowfall, wind speed, solar radiation, and ET_0 (FAO Penman-Monteith reference evapotranspiration, pre-calculated by the API).	**Requests** fetches the JSON API. **BeautifulSoup** parses the IMGW station list HTML page for metadata.
Meteostat	Station metadata: name, WMO/ICAO identifiers, elevation, coordinates, timezone, nearby stations (all JS-rendered content).	**Selenium** loads the JS-rendered page. **BeautifulSoup** + **regex** extract structured data from rendered HTML.

The merged DataFrame will contain ~462 rows (daily) and ~12 columns. A cumulative water balance chart will visualize surplus/deficit periods to guide garden irrigation decisions.